

ADEN GUO

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TECHNICAL SKILLS

- **Languages & Data:** Python, TypeScript, JavaScript, Go, Ruby, Rust, C++, SQL, NumPy, Pandas, scikit-learn, PyTorch, OpenCV
- **Web & Frameworks:** React, Next.js (App Router), SvelteKit, Node.js, FastAPI, Ruby on Rails, Go + go-app (WebAssembly), PWA / Service Workers, Tailwind CSS, Tauri, Drizzle ORM, Bun
- **Databases & Retrieval:** PostgreSQL, pgvector (hybrid semantic search), SQLite, Redis, Supabase, Neon
- **Cloud & Infrastructure:** Azure, Azure DevOps (Repos & Pipelines), AWS S3, Vercel, Railway, Docker, CI/CD pipelines, Distributed & event-driven systems
- **AI & LLM Engineering:** Anthropic / Claude API, OpenAI, Vercel AI SDK, OpenRouter, RAG & vector retrieval, Prompt caching, MCP & agentic CLI tooling
- **Systems & DevOps:** Linux, Bash, Git & GitHub, Docker, Playwright, Web Push (VAPID), RSpec / pytest, VS Code
- **GIS & Survey:** ArcGIS, QGIS, PostGIS, GeoPandas, GDAL/PROJ, RTK GNSS workflows, Agisoft Metashape

PROFESSIONAL EXPERIENCE

GIS Technologist - Canada West Land, Calgary, AB

Dec 2025 - Present

- Manage six production GIS maps for a major enterprise client and support data visualization workflows across feature, land, and reference datasets ranging from roughly 5,000 to 1M records.
- Built end-to-end ETL and data-cleaning pipelines with Esri, FME, Python scripting, APIs, and HTML/CSS expression styling for polished professional map outputs.
- Contributed to an internal Azure-hosted production app for agentic AI workflows using LangChain, LangGraph, and n8n, with role-specific system-prompt design; reduced selected full-day manual processes to 1-2 hours.
- Developed and shipped within an Azure DevOps environment — source control and code review through Azure Repos, automated build and release via Azure Pipelines (CI/CD), and secrets managed with Doppler — collaborating in a branch-based workflow on the internal production application.

Engineering Capstone Project Lead - University of Calgary, Calgary, AB

Sep 2024 - Apr 2025

- Led capstone delivery for a geospatial mapping study, comparing survey methods and validating spatial datasets with photogrammetry tools and Python-based analysis.

Shift Lead / Team Trainer - Gongcha

Nov 2020 - May 2024

- Managed high-volume shift operations and trained staff on standardized workflows, strengthening team coordination and execution under time constraints.

EDUCATION

Bachelor of Science: Geomatics Engineering

University of Calgary - Calgary, AB

2020 - 2026

Relevant Skills:

- Software-relevant foundation in spatial databases, data analysis, Python workflows, and technical reporting

PROJECTS

Glass Atlas — Source-Grounded RAG Knowledge Site - glass-atlas-production.up.railway.app

- Deployed SvelteKit 2 (Svelte 5 runes) editorial site that turns a personal note library into a conversational, citation-backed assistant.
- Engineered hybrid semantic + lexical retrieval over Neon PostgreSQL with pgvector (cosine similarity) and Drizzle ORM, with confidence-tiered fallbacks that keep every answer grounded in source notes.
- Built streaming RAG chat through OpenRouter (Gemini 2.0 Flash) on text-embedding-3-small 1536-dim vectors, with strict-TypeScript end-to-end typing.
- Shipped on Railway with a Bun HTTP server, Auth.js GitHub OAuth, and private S3-compatible storage via presigned URLs.

Weather & Wellness — UBC Research Platform - ubcpsych.com

- Production multi-tenant research platform for a UBC Psychology lab where research assistants run validated psychological instruments with anonymous participants.
- Built a Next.js 15 / React 19 frontend and FastAPI backend that auto-scores validated instruments (digit span, loneliness, depression, anxiety) in real time.
- Implemented weather-psychology mixed-effects models with statsmodels, NumPy, and Pandas over per-lab isolated datasets.
- Architected per-lab data isolation on Supabase Postgres/Auth with Upstash Redis, shipped to production at ubcpsych.com.

PigeonCoop — Local-First Agent Workflow IDE - github.com/AG9898/PigeonCoop

- Active-development Rust/Tauri desktop app for designing, running, monitoring, and replaying agent workflows against local repositories.
- Built a React/TypeScript graph workspace with Start, End, Agent, Tool, Router, Memory, and Human Review nodes, plus pre-run validation and workspace-scoped CLI execution for repository tasks.
- Designed Rust core crates for typed workflow definitions, RunEvent envelopes, deterministic run/node state machines, runtime adapters, and SQLite-backed workflow/run/event persistence.
- Added replay/debugging and human-review surfaces backed by Rust unit tests, React component tests, and WebdriverIO/Tauri E2E coverage across builder, live run, review, and replay flows.